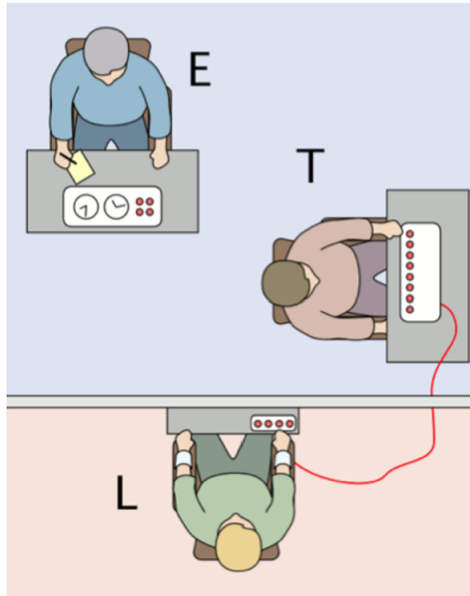


Lecture 13: The Geometric Distribution

Running example: the Milgram experiment



The Bernoulli Distribution

Example: Roll a fair six-sided die and consider the roll a “success” if a 2 or a 3 comes up.

The Geometric Distribution

Dr. Mario wants to repeat Milgram’s experiment, but he only wants to sample people until he finds someone who will not inflict a severe shock. What is the probability that he stops after ...

- ... the first person?
- ... the third person?
- ... the tenth person?

The geometric distribution describes ...

Possible values and probabilities:

Example: Probability that it will take 6 rolls of a fair six-sided die to produce the first 6?

Expected Value and Variance for a Geometric Random Variable

Example: back to Milgram experiment

Example: back to rolling a 6 with a fair die

More interesting example: Suppose there is an 85% chance it will not rain on any given day, independent of all other days.

- Given that it rains today, what is the probability that it will rain again tomorrow?
- Calculate the mean and standard deviation for the number of days until we see rain again.
- Which future day (relative to today) is most likely to be the next day we see rain?